sign constituent

$$W = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} \qquad P_{t,2} = \begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix} \qquad \mathrm{Tr}(PW) = 3$$

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standard constituent

$$W = \begin{pmatrix} 6 & 4 \\ 3 & 1 \end{pmatrix} \qquad P_{s,1} = \begin{pmatrix} 0 & 4 \\ 0 & 1 \end{pmatrix} \qquad \mathrm{Tr}(PW) = 6$$